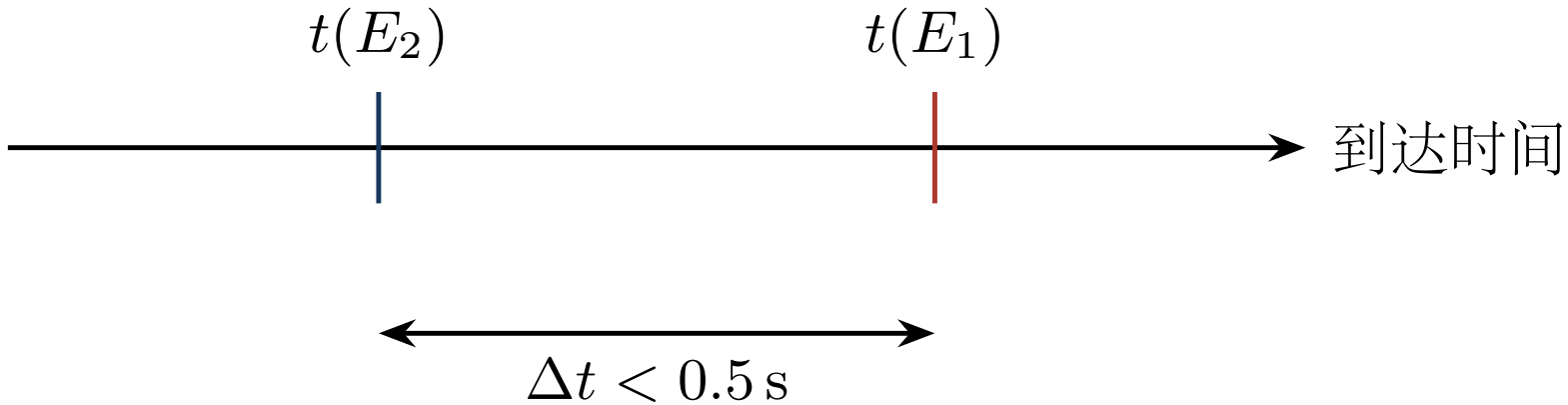
$$D = 7.5 \times 10^9 \text{ 光年}$$

$$E_1 = 0.1\,\mathrm{MeV},\quad E_2 = 100\,\mathrm{MeV}$$

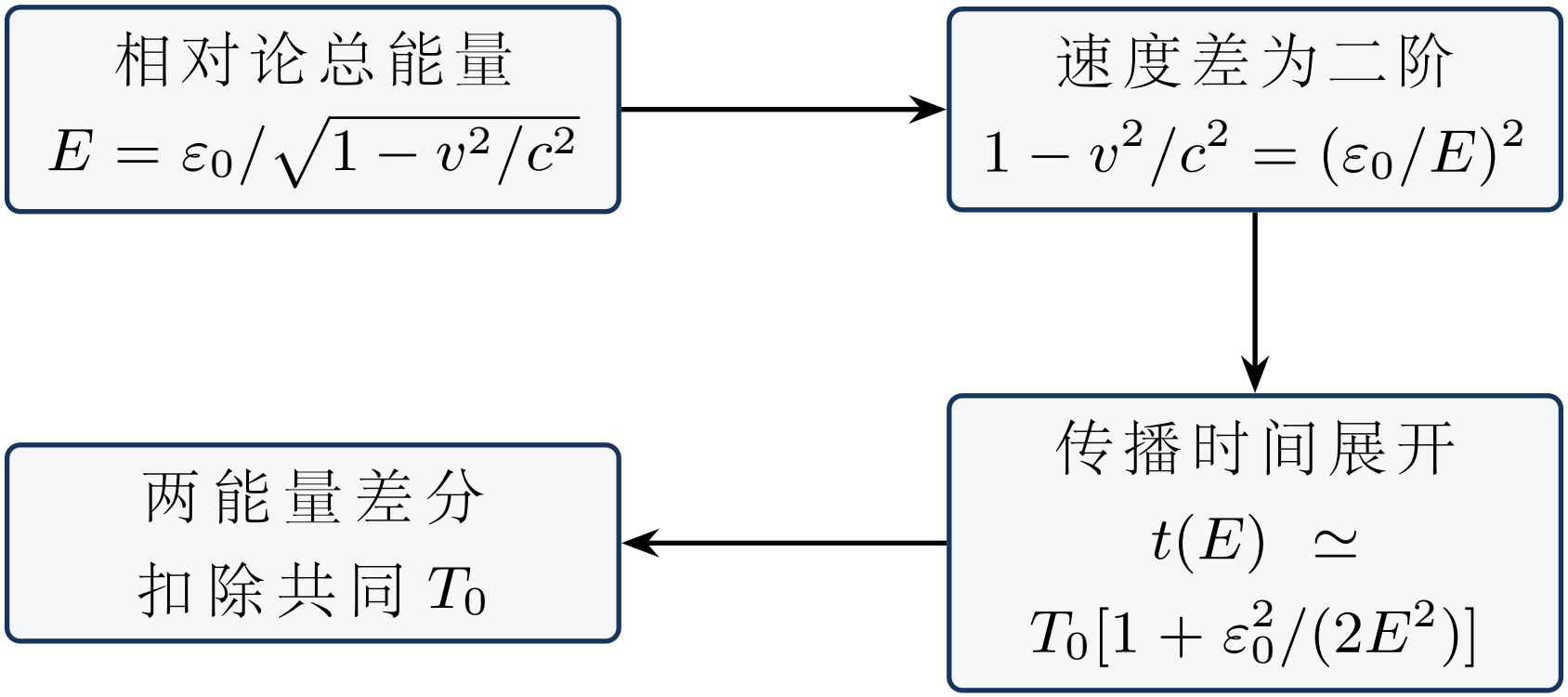
$$\Delta t < 0.5\,\mathrm{s};\ \text{取 } c = 2.998 \times 10^8\,\mathrm{m/s}$$

按给定平直传播模型估算静止能量上限

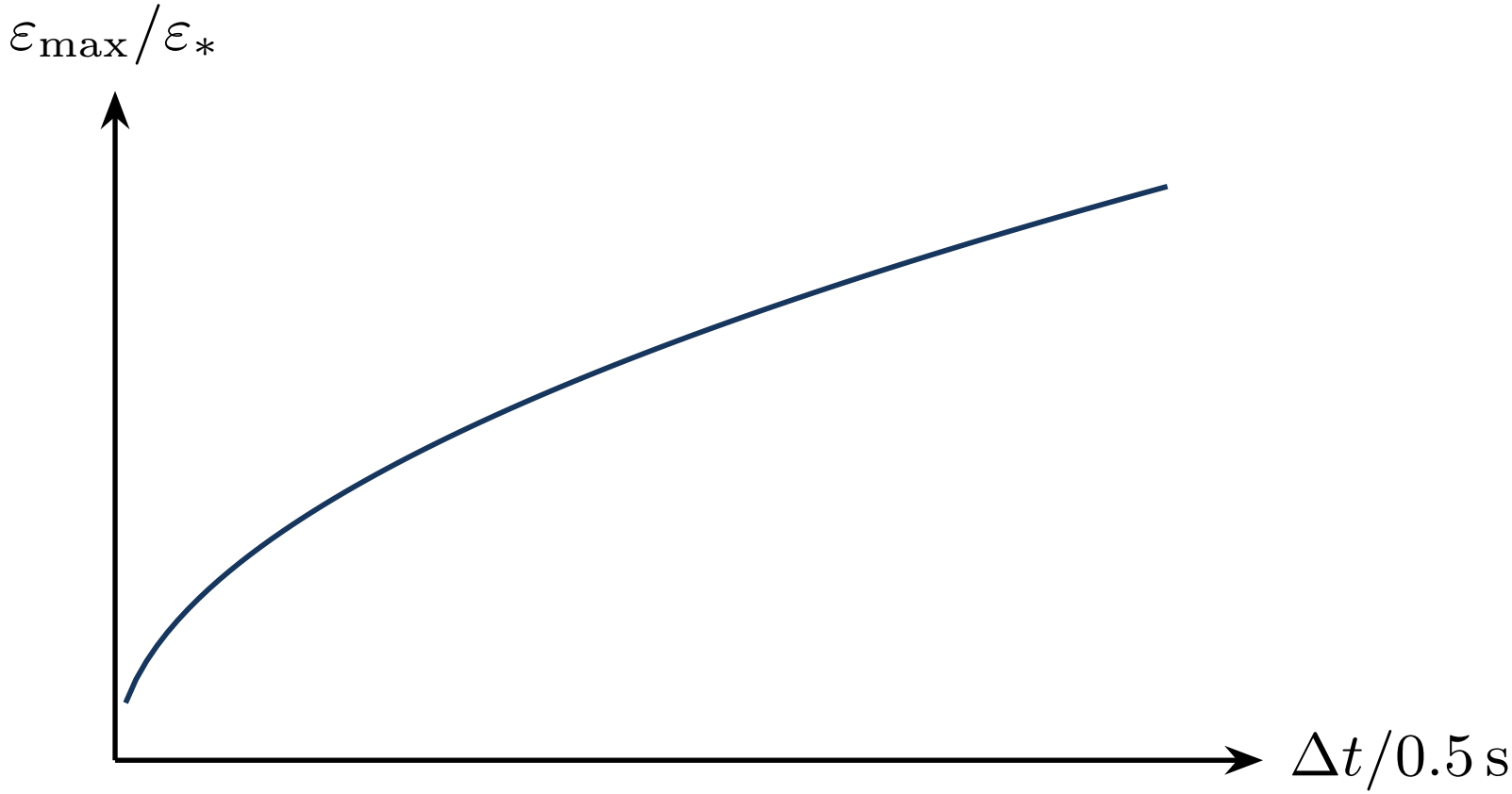


共同传播时间约 $2.36682 \times 10^{17}\,\mathrm{s}$

计时图非按比例；共同出发假设决定质量约束

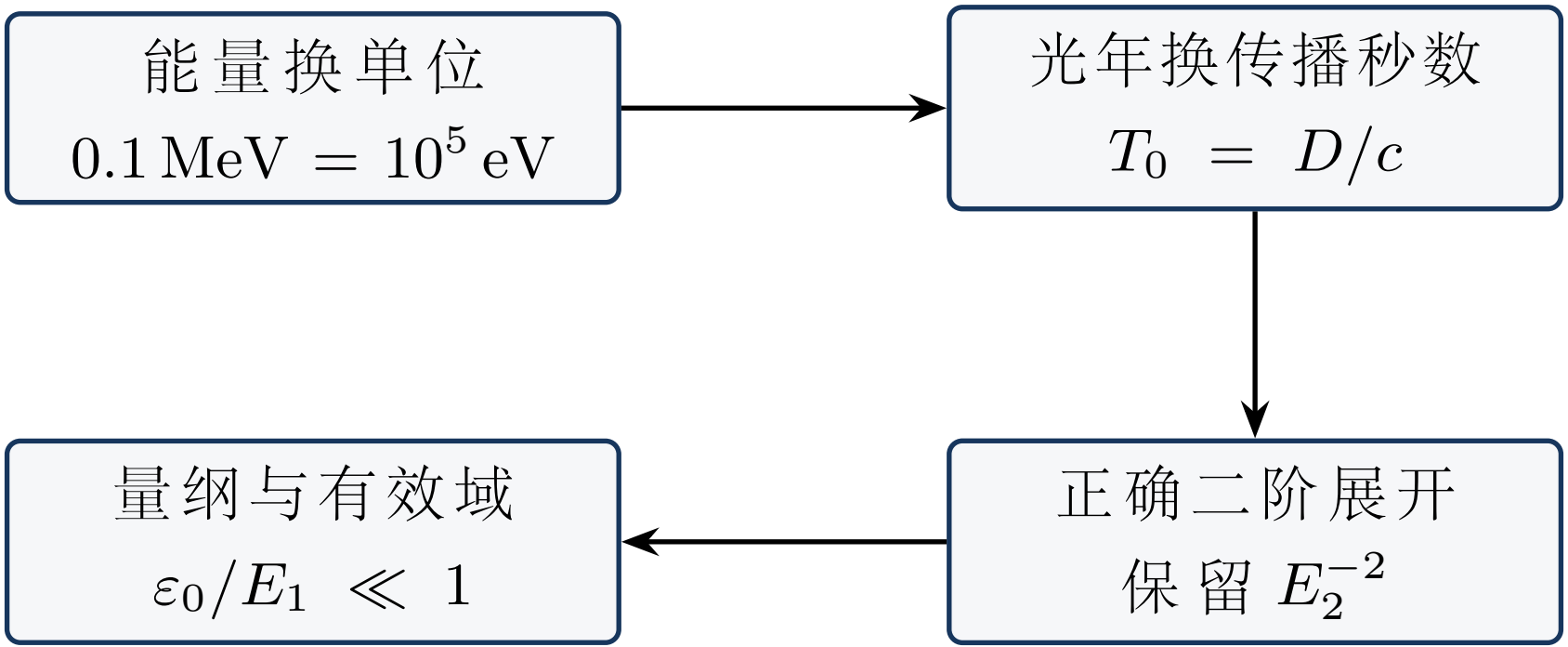


$\varepsilon_0 = m_0c^2$；平方不可漏掉



$$\varepsilon_* \simeq 2.056 \times 10^{-4}\,\mathrm{eV}$$

上限随 $\sqrt{\Delta t}$ 增长，随 $1/\sqrt{D}$ 减小



这是题设条件下的估算，不宣称为现行最强实验上限